SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address: 192.168.10.65**
- **Subnet Mask: 255.255.255.192 (/26)**
- **Default Gateway: 192.168.10.1**

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- **Identifying the subnet to which the configured IP address belongs.**
- **Determining the network address, broadcast address, and valid host range.**
- **Verifying whether the configured IP address and gateway are valid.**
- **Identifying the configuration errors.**
- **Optimizing the configuration by assigning an appropriate IP address and default gateway.**
- **Calculating the total number of usable host addresses after correction.**

Given Configuration

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

1. Identify the subnet to which the configured IP belongs

A **/26** subnet has a block size of **64**.

Subnets are:

Subnet	**Address Range**
192.168.10.0/26	192.168.10.0 – 192.168.10.63
192.168.10.64/26	192.168.10.64 – 192.168.10.127
192.168.10.128/26	192.168.10.128 – 192.168.10.191
192.168.10.192/26	192.168.10.192 – 192.168.10.255

The IP **192.168.10.65** belongs to **192.168.10.64/26**.

Explanation:

- A **/26 subnet mask (255.255.255.192)** creates subnets of 64 IP addresses each.
- The workstation IP falls within the second subnet, not the first.
- Identifying the correct subnet helps determine whether devices can communicate directly.

2. Determine the network address, broadcast address and valid host range

Item	**Value**
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Valid Host Range	192.168.10.65 – 192.168.10.126

Explanation:

- The **network address** identifies the subnet and cannot be assigned to a host.
- The **broadcast address** is used to send data to all devices within the subnet.
- Only addresses between the network and broadcast addresses are valid for hosts.

3. Verify whether the configured IP address and gateway are valid

IP Address

- 192.168.10.65 is the first usable host.
- Therefore, the IP address is **valid**.

Default Gateway

Gateway = **192.168.10.1**

This belongs to subnet **192.168.10.0/26**, while the workstation belongs to **192.168.10.64/26**.

Therefore, the gateway is **invalid**.

Explanation:

- The workstation IP (192.168.10.65) belongs to the subnet **192.168.10.64/26**.
- The configured gateway (192.168.10.1) belongs to the subnet **192.168.10.0/26**.
- A host and its default gateway must always belong to the **same subnet** for proper communication.

4. Identify the configuration errors

- Workstation IP address is correct.
- Default gateway is configured in the wrong subnet.
- Devices in different subnets cannot communicate without proper routing.

5. Optimize the configuration

Parameter	Correct Value
IP Address	192.168.10.65
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.66 (or any valid router interface in the same subnet)

Explanation:

- The router interface should use an address within the same subnet.
- Assigning the gateway as **192.168.10.66** allows successful communication.
- No changes to the IP address are required because it is already valid.

6. Calculate the total usable host addresses

Formula:

$$\text{Usable Hosts} = 2^{(32-26)} - 2$$

$= 2^6 - 2$

$= 64 - 2$

= **62 usable hosts**

Explanation:

- Two addresses are reserved in every IPv4 subnet.
- One is the network address and the other is the broadcast address.
- Therefore, only **62 addresses** are available for hosts.

Final Answer

Requirement	**Answer**
Subnet	192.168.10.64/26
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Host Range	192.168.10.65–192.168.10.126
IP Valid?	Yes
Gateway Valid?	No
Configuration Error	Gateway belongs to another subnet
Optimized Gateway	192.168.10.66
Usable Hosts	62

2. Two devices are configured with the following IPv6 addresses:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- **Expanding both IPv6 addresses into their complete notation.**
- **Determining the network prefix of each device.**

- **Verifying whether both devices belong to the same subnet.**
- **Identifying the addressing or prefix mismatch.**
- **Optimizing the IPv6 configuration by assigning appropriate addresses.**
- **Calculating the number of available host addresses within the corrected subnet.**
-

Given

Device A **2001:db8::ff00:42:8329/64**

Device B **2001:db8:0:0:1::1/64**

1. Expand both IPv6 addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Explanation:

- IPv6 allows leading zeros to be omitted and consecutive zero groups to be compressed using "::".
- Expanding the addresses makes it easier to compare all 128 bits.
- This helps identify the correct network prefix.

2. Determine the network prefix

Since both addresses use **/64**, the first four hexadecimal groups represent the network.

Device A Network Prefix

2001:db8:0:0::/64

Device B Network Prefix

2001:db8:0:0::/64

Explanation:

- A **/64 prefix** means the first 64 bits identify the network.
- Devices with the same first 64 bits belong to the same subnet.

- Both devices satisfy this condition.

3. Verify whether both devices belong to the same subnet

Both devices have the same network prefix:

2001:db8:0:0::/64

Therefore, **both devices belong to the same subnet**.

Explanation:

- Since both devices share the same network prefix, they should communicate directly.
- If communication fails, the problem is likely due to routing, interface configuration, or gateway settings rather than the IPv6 address.

4. Identify the addressing or prefix mismatch

There is no IPv6 addressing or prefix mismatch.

Communication failure may be caused by:

- Incorrect default gateway
- Interface shutdown
- Duplicate IPv6 address
- Missing IPv6 routing

5. Optimize the IPv6 configuration

Device	IPv6 Address
Device A	2001:db8:0:0::10/64
Device B	2001:db8:0:0::20/64
Gateway	2001:db8:0:0::1

Explanation:

- Assigning simple sequential addresses improves readability and troubleshooting.
- All devices remain within the same /64 subnet.

6. Calculate available host addresses

Host bits = **128 − 64 = 64**

Available addresses

= $\mathbf{2^{64}}$

= **18,446,744,073,709,551,616**

Final Answer

Requirement	**Answer**
Expanded Address A	2001:0db8:0000:0000:0000:ff00:0042:8329
Expanded Address B	2001:0db8:0000:0000:0001:0000:0000:0001
Network Prefix	2001:db8:0:0::/64
Same Subnet?	Yes
Mismatch?	None
Optimized Addresses	::10 and ::20
Available Hosts	2^{64}

3. An organization has allocated the network 192.168.1.0/24 **into several departments using the subnet masks** /26, /27, **and** /28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.**

<u>Task</u>

Analyze and optimize the subnet allocation by:

- **Identifying the network address, broadcast address, and host range for each subnet.**
- **Calculating the number of usable and unused IP addresses.**
- **Identifying inefficient subnet allocations.**
- **Debugging the addressing plan by locating subnet allocation mistakes.**
- **Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.**

Given

Network: **192.168.1.0/24**

Allocated subnets:

- /26
- /27

- /28

1. Identify network address, broadcast address and host range

Subnet	Network	Broadcast	Host Range
/26	192.168.1.0	192.168.1.63	192.168.1.1–192.168.1.62
/27	192.168.1.64	192.168.1.95	192.168.1.65–192.168.1.94
/28	192.168.1.96	192.168.1.111	192.168.1.97–192.168.1.110

2. Calculate usable and unused IP addresses

Subnet	Total Addresses	Usable Hosts
/26	64	62
/27	32	30
/28	16	14

Example departmental requirement:

Department	Required	Allocated	Unused
Dept A	50	62	12
Dept B	20	30	10
Dept C	10	14	4

Total unused inside allocated subnets

= **12 + 10 + 4 = 26**

Entire /24 usable hosts

= **254**

Allocated usable hosts

= **62 + 30 + 14 = 106**

Remaining usable addresses

= **254 − 106 = 148**

3. Identify inefficient subnet allocations

Problems include:

- Small department allocated a large subnet.
- Large department allocated a small subnet.
- Many IP addresses remain unused.

4. Debug the addressing plan

Possible mistakes:

- Incorrect subnet size assigned.
- Overlapping subnet ranges.
- Wrong gateway configuration.
- Departments exceeding subnet capacity.

5. Recommend optimized VLSM

Department	Required Hosts	Recommended Subnet
Dept A	50	192.168.1.0/26
Dept B	20	192.168.1.64/27
Dept C	10	192.168.1.96/28

This VLSM allocation minimizes address wastage while meeting host requirements.

Explanation:

- VLSM assigns subnet sizes according to the number of required hosts.
- This minimizes unused addresses while allowing future expansion.

Final Answer

Requirement	Answer
Network Details	Shown above
Usable Hosts	62, 30, 14
Unused in Example	26
Remaining in /24	148
Problems	Incorrect subnet allocation and wasted IPs
Optimization	Use VLSM

4. **Two devices are configured with the following IPv6 addresses:**

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

<u>Task</u>

Debug the IPv6 configuration by:

- **Expanding both IPv6 addresses into their complete notation.**
- **Determining the network prefix of each device.**
- **Verifying whether both devices belong to the same subnet.**
- **Identifying the addressing or prefix mismatch.**
- **Optimizing the IPv6 configuration by assigning appropriate addresses.**
- **Calculating the number of available host addresses within the corrected subnet.**

Given:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, they are unable to communicate.

1. Expand both IPv6 addresses into their complete notation

Device A

Compressed Address: 2001:db8::ff00:42:8329/64

Expanded Address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

Compressed Address: 2001:db8:0:0:1::1/64

Expanded Address: 2001:0db8:0000:0000:0001:0000:0000:0001

Explanation:

- IPv6 uses **"::"** to replace one or more consecutive groups of zeros.
- Expanding the addresses into eight groups of four hexadecimal digits makes it easier to compare the network portions and identify configuration errors.

2. Determine the network prefix of each device

Since both devices use a /64 prefix, the first 64 bits (first four hexadecimal groups) represent the network.

Device	**Network Prefix**
Device A	2001:db8:0:0::/64
Device B	2001:db8:0:0::/64

Explanation:

- A /64 prefix means the first four groups identify the network, while the remaining four groups identify the host.
- Both devices have the same first four groups (2001:db8:0000:0000).

3. Verify whether both devices belong to the same subnet

Yes.

Both devices belong to the subnet:

2001:db8:0:0::/64

Explanation:

- Since both devices share the same network prefix, they are members of the same IPv6 subnet.
- Normally, devices in the same subnet should communicate directly without requiring a router.

4. Identify the addressing or prefix mismatch

There is no IPv6 addressing or prefix mismatch.

Both devices are correctly configured with the same /64 network prefix.

Therefore, the communication problem is likely caused by another configuration issue, such as:

- Incorrect or missing default gateway
- Network interface is disabled
- IPv6 is disabled on one device
- Firewall blocking IPv6 traffic
- Switch or router configuration issue

Explanation:

- The IPv6 addresses themselves are valid.
- Since there is no prefix mismatch, troubleshooting should focus on the network configuration rather than the addressing scheme.

5. Optimize the IPv6 configuration

Use simple sequential addresses within the same subnet.

Device	**Optimized IPv6 Address**
Device A	2001:db8:0:0::10/64
Device B	2001:db8:0:0::20/64
Default Gateway	2001:db8:0:0::1/64

Explanation:

- Assigning sequential addresses makes network management and troubleshooting easier.
- All devices remain within the same 2001:db8:0:0::/64 subnet, ensuring proper communication.

6. Calculate the number of available host addresses within the corrected subnet

Prefix Length = **/64**

Host Bits = **128 − 64 = 64**

Formula:

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$$

Available Host Addresses = 18,446,744,073,709,551,616

Explanation:

- IPv6 provides a very large address space compared to IPv4.
- A /64 subnet is the standard subnet size used in IPv6 and supports approximately 18.4 quintillion interface addresses.

Final Answer

Task	Answer
Expanded Address (Device A)	2001:0db8:0000:0000:0000:ff00:0042:8329
Expanded Address (Device B)	2001:0db8:0000:0000:0001:0000:0000:0001
Network Prefix (Both Devices)	2001:db8:0:0::/64
Same Subnet?	Yes
Addressing/Prefix Mismatch	No mismatch found; both devices share the same /64 network prefix.
Optimized IPv6 Addresses	Device A → 2001:db8:0:0::10/64 Device B → 2001:db8:0:0::20/64 Gateway → 2001:db8:0:0::1/64
Available Host Addresses	2^{64} = 18,446,744,073,709,551,616

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	**10.0.0.2**
192.168.2.0/24	**10.0.0.3**
Default Route	**10.0.0.1**

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- **Identifying the matching routing entry using the longest-prefix match.**
- **Determining the next-hop router selected for packet forwarding.**
- **Verifying whether the destination IP belongs to the intended subnet.**
- **Identifying any routing configuration errors.**
- **Optimizing the routing table to improve packet forwarding efficiency.**
- **Identifying the route that will be used if no matching destination exists.**

Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP: **192.168.2.100**

1. Identify the matching routing entry using longest-prefix match

The destination 192.168.2.100 belongs to 192.168.2.0/24.

Matching route:

192.168.2.0/24 → 10.0.0.3

Explanation:

- Routers always choose the route with the longest matching prefix.
- In this case, the /24 route exactly matches the destination network.

2. Determine the next-hop router

Next hop selected: **10.0.0.3**

Explanation:

- The next-hop router is the device that forwards the packet toward its destination.
- The router sends the packet to 10.0.0.3 before it reaches the destination network.

3. Verify whether destination belongs to intended subnet

Subnet: 192.168.2.0/24

Host range: 192.168.2.1 – 192.168.2.254

Destination: 192.168.2.100

Therefore, the destination belongs to the intended subnet.

Explanation:

- The destination IP falls within the valid host range.
- Therefore, it belongs to the 192.168.2.0/24 network.

4. Identify routing configuration errors

Possible errors:

- Incorrect next-hop address.
- Interface connected to 10.0.0.3 is down.
- Static route missing or incorrect.
- No return route from the destination network.

5. Optimize the routing table

Recommended improvements:

- Verify reachability of next-hop **10.0.0.3**.
- Correct any incorrect static routes.
- Add missing return routes.
- Use dynamic routing protocols (OSPF, EIGRP, or RIP) for better scalability.

6. Identify the route used if no matching destination exists

If no specific route matches, the router uses the **default route (0.0.0.0/0)**.

Next hop: 10.0.0.1

Default Route

0.0.0.0/0 → **10.0.0.1**

Explanation:

- If no specific route matches the destination, the router forwards the packet using the default route.
- This ensures packets for unknown networks are still forwarded toward another router or the internet.

Final Answer

Requirement	Answer
Matching Route	192.168.2.0/24
Longest Prefix	/24
Next Hop	10.0.0.3
Destination Valid?	Yes
Possible Errors	Wrong next hop, interface down, missing return route
Optimization	Verify next hop and use dynamic routing if needed
Default Route Used	10.0.0.1

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation